Technology Stack

Date	25 June 2026
Team ID	xxxxxx
Project Name	AI-Crop Reccomendation System
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Technology Stack Details:

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	**Frontend / Client-Side** (e.g., React, Angular, HTML/CSS)	HTML5, CSS3, JavaScript, Bootstrap	These technologies are used to develop a responsive, interactive, and user-friendly web interface. Bootstrap provides a mobile-friendly design and reusable UI components, while HTML, CSS, and JavaScript.

2	**Backend / Server-Side** (e.g., Node.js, Python/Django, Java)	Python (Flask)	Flask is a lightweight and efficient Python web framework that enables quick development of web applications. It integrates seamlessly with machine learning models, simplifies API development.
3	**Database / Data Storage** (e.g., MySQL, MongoDB, PostgreSQL)	MySQL	MySQL is a reliable relational database management system used to securely store applicant information, loan details, and prediction results. It supports efficient data retrieval, maintains data integrity, and provides fast query execution for handling structured data.
4	**Cloud / Hosting / Deployment** (e.g., AWS, Azure, Heroku, Docker)	Render	Render provides a simple and cost-effective cloud platform for deploying Flask applications. It supports continuous deployment from GitHub, ensures reliable application hosting, offers automatic updates, and allows users to access the application.

5	**Version Control & CI/CD** (e.g., GitHub, GitLab, Jenkins)	GitHub	GitHub is used for version control, source code management, and collaboration among team members. It tracks code changes, maintains version history, enables secure cloud backup, simplifies project management.
6	**Third-Party APIs/ Other Tools** (e.g., Stripe, Twilio, Google Maps)	Google Colab, Kaggle Loan Prediction Dataset	Google Colab provides a cloudbased environment for developing, training, and testing machine learning models without requiring powerful local hardware. The Kaggle Loan Prediction Dataset supplies high-quality real-world data for preprocessing, and reliability of the loan approval prediction system.